🌴 Plants

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Overview

- [ ] Hero Plants are environment object placed by the player.
- [ ] Hero Plants can propagate (e.g., fern), spawning more plants around. See propagation.
- [ ] Can grow/upgrade and change model. See growing/upgrading.
  - [ ] Growing rate can be accelerated by spending dino points.
  - [ ] Growing rate can be accelerated by being near fungi.
  - [ ] Acceleration near fungi consumes Nitrogen from Fungi
- [ ] Generates Nutrients points over time See nutrient points generation.
- [ ] Nutrient points can be Eaten by Dinosaurs.
- [ ] Has a ‘Cover’ Value that changes based on Growth Stage and proximity to other plants. See Cover.
  - [ ] Can be grouped into a rest area, dinosaurs use rest areas to sleep. Changes according to growth.
- [ ] Can Die if their nutrient value hits zero.
- [ ] Can Regrow X number of times.
- [ ] Growth and Regrowth can be altered by Growth Condition. See growing/upgrading.
- [ ] They are purchased using Dino Points and planted using the planter.
- [ ] Have a detection radius which is used for propagation, to detect likes and dislikes and to establish a viable range for creating rest areas.

- [ ] Plants have a configurable ‘barrier’ radius in which other plants cannot be placed or propagate.
- [ ] Plants are unlocked as the player levels up

Unlocking

Two plants will be unlocked by the tutorial - the Fern and the Fungi.

When the tutorial is complete, the plants unlock and are economized by points value.

The most useful and nutrient-rich plants should be highest cost.

When the tutorial is complete, all plants are available to the player in the **Inventory** (as opposed to unlock gradually by player level (though we want to keep that option for balancing purposes)).

Planting & Status

Max cap

> ⚠ Due to performance, there will be a cap for how many plants the player can plant in the environment. This can be used as a *game mechanic* where these limitiations force the player to be more mindful of what plants to place in the environment.
>
> For example, we can make it so that planting just ferns up to max limit (say 20 plants) will not be enough to feed 6 adult dinosaurs.
>
> This max plant cap needs to be visualized in the inventory (e.g. bottom left) like so: 🌿**5/20**
>
> *But first we need to know what the actual hardware limits are, before we start designing around them.*

Mechanics

Hero Plants are placed by the player using the planter.

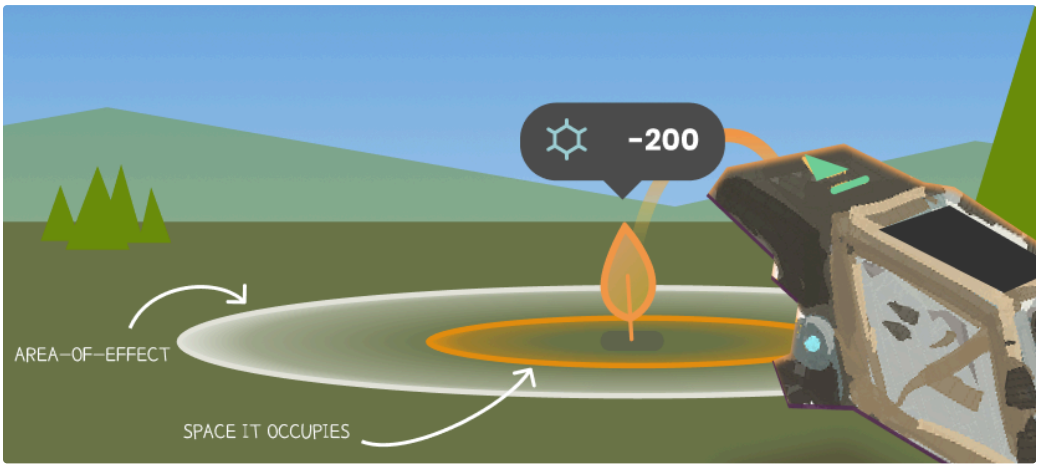
> ℹ An exception is the tutorial in which designers place ‘starter’ plants and the player must scan these to unlock the ‘Fern’ plant type.

Hero Plants cost Dino Points to plant - this is called the **Plant Cost DP** . When plants are removed, a percentage of these are reclaimed and this is declared as the **Plant Removal DP**. Both these sums are visible as a tooltip above the target.

Plants have a detection radius (area-of-effect). This is visualized when “planting” as a circle around the plants.

The player can also review this radius by scanning a plant after it is already planted. The size of this radius is determined by whether or not the plant is a “Parent” or “Propagated” plant.

Plants have space they exclusively occupy. This prevents the player from planting plants ontop of each other, and make sure they don’t clip/overlap with other plants or the terrain as they grow from a seedling to a tree.



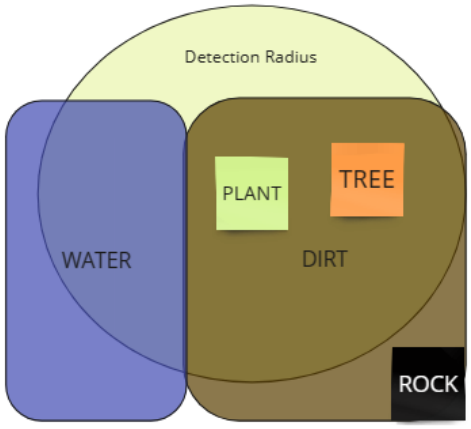
Plants have a configurable barrier radius inside which other plants cannot be placed.

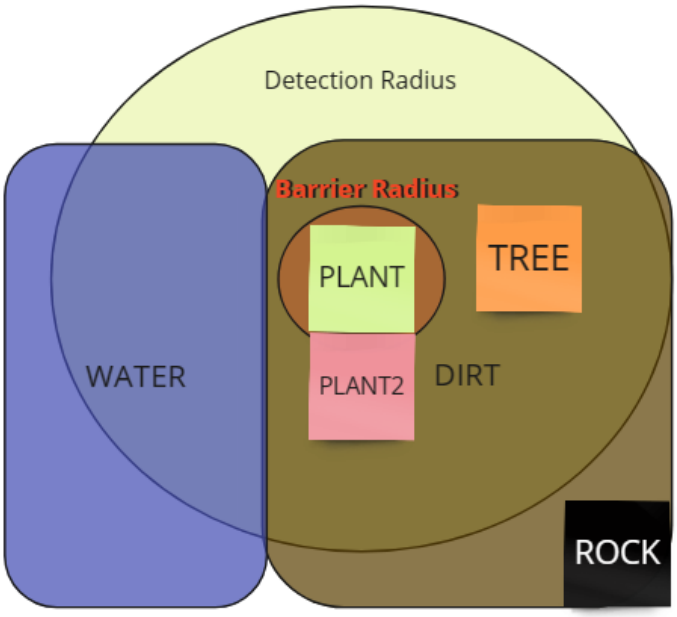
Visually when plants are planted, their rotation is randomized to give visual variation to the plants.

Hero Plants placed by the player have an **green glow** when the scanner is held. Environmental plants have no glow and are not interactable.

Plants that are freshly planted cannot yet propagate; this is called the **Propagation Interval**.

Plants also have a Parent or Propagated status - this is a boolean called “**IsParent**”. This detemines which detection radius plants use to determine their likes and dislikes. Plants placed by the player are parents, plants propagated from that plant are not parent plants.





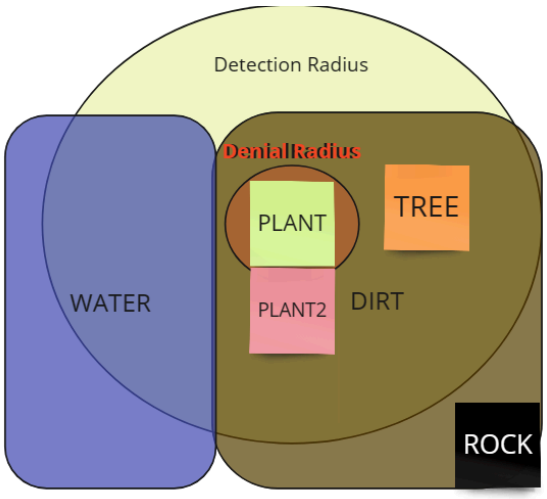
Variable	Explanation	Example Value
Plant Cost DP	The amount of Dino Points it costs for a player to plant a plant from the inventory.	
Plant Removal DP	The residual amount of **Dino Points** the player receives for removing a plant. A percentage of the original cost out of a possible 100.	
Parent Detection Radius	The size of the collider on plants planted by	

	the player. This is used to detect likes, dislikes, and proximity.	
Propagated Detection Radius	The size of the collider on plants propagated by other plants. This is used to detect likes, dislikes, and proximity. It is intended for this to be smaller than parent plants.	
Propagation Interval	is the interval in hours before a new plant can spawn from a single parent plant.	
IsParent	Whether or not the plant is a Parent or Propagated plant. Used to determine other factors like the detection radius.	0-1

Placement

Plants have a collision ‘denial radius’.
This radius prevents the placement of other plants inside of the radius, and is configurable by design.

The recommendation is that all plants have a collision radius that is 75% of the size of the total plant.
In most cases this will mean the leaves can overlap slightly but no trunks or bodies of the bushes should collide, and this can be adjusted on an individual basis.



The holographic representation will turn red when it collides with the denial radius of an existing plant and will not allow the player to plant. A denial squelch will play and the multitool will flash 'INVALID' on it's screen.

Plants have a maximum angle value that determines how steep a surface they can be placed on.

Whether or not plants can be placed in water is also configurable on a basis of 0-1. Shallow water such as a puddle or pond and deep water such as an ocean or river are differentiated.

Scale and rotation can also be randomized to increase visual diversity.

Note: A balanacing pass will be required to ensure that all detection radii are large enough to function with the barriers or else there is a high risk the game will suddenly stop functioning if (for example) cover detection no longer works because plants must be farther apart.

Variable	Explanation	Example
Max Angle	The steepest angle a given plant can be placed on.	60
Denial Radius	The volume inside which no other plants can be planted on a	225

	given plant. To be 75% of the size of the stage 3 plant to start, for all plants. E.G. if a fern is 300 Unreal Units large then the denial radius of all stages of fern is 225.	
DeepWater	Can a plant be placed in deep water?	0 or 1
ShallowWater	Can a plant be placed in shallow water?	0 or 1
RandomizeRotation	Whether or not a plants rotation is randomized on placement.	0 or 1
ScaleRandomization	How much the scale of a given plant is randomized when placed.	0 to 1, as a percentage out of 100. E.G. 0.05 = up to 5% randomized deviation in scale.

Nutrients

Plants have nutrient value. This represents the amount of 'food' in the plant, but also the plants overall health. The bigger the plant, the more nutrients. Nutrients are visualized through changes to the plant model itself.

BUSHES - as bushes surpasses certain 'caps' defined in the DT_Plantdata data table, the model changes to reflect its new growth and increased value.

TREES - Trees do not de-grow when they pass their cap. Instead, the leaves of the tree are grazed away and re-grow. Trees that bear fruit drop it to the ground for babies to consume and this is also represented visually.

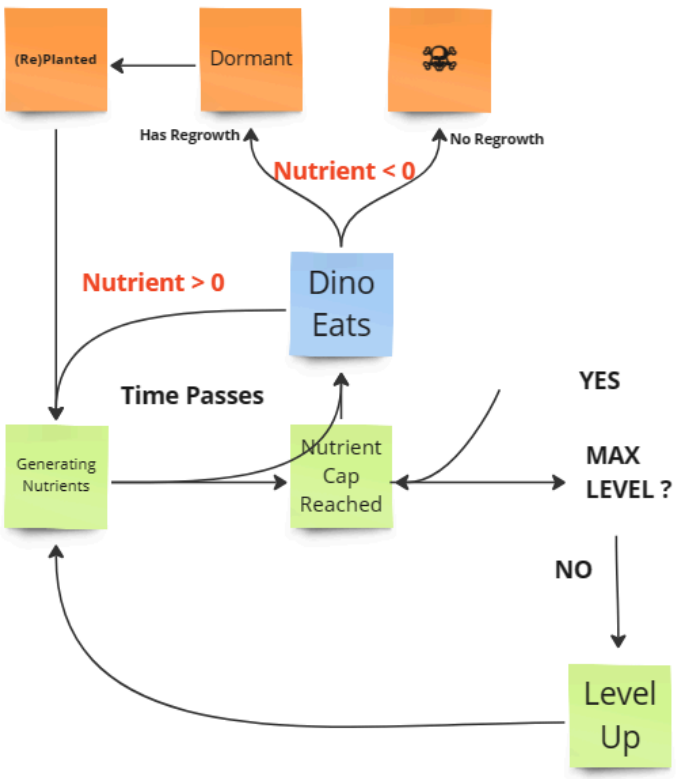
Nutrients can be compared to the needed consumption of the Herd (Herd should be a page if it isn't) to allow the player to visualize needs of the group.

Other visual ideas to show plant nutrients is that plants will sparkle and show other particles (insects etc.) when they have more nutrient points and slowly lose their sparkles as they lose the nutrients. Plants will also have a sparkle (e.g., a different coloured one distinct from nutrients) when they have a cover value of at least 3.

The sparkles will be drained when Dinos eat them. When scanned the saturation of the plants **green glow** (as opposed to orange of dinosaurs) will be brighter when a plant's nutrients are closer to max and duller when nutrients are closer to 0. Lastly, as plants generate nutrients, nutrient point bubbles (similar to the Dino Point bubbles) will be seen traveling up to the Harmony Meter.

These points can be visualized by scanning the plants using the scanner.
The data that is scanned will populate the DinoDex.
The player receives periodic updates about the points being generated by their plants, these are visualized on the UI.

If plants are not producing enough resources to support the dinosaur population, the player should see the issue reflected in the health of their dinosaurs and the appearance (or lack) of the plants.



Basic Lifecycle of a Plant

Variable / Condition	Explanation:		Example Value
Base Nutrient Generation Rate	float (/minute), how many nutrients are generated every Nutrient tick.	The time interval at which Nutrient Points are produced.	1
Nutrition Generation Rate	The nutrient generation rate after modifier is applied.		
Compost Decrement	The amount of compost the		

	plant takes from the fungi per tick		
Nutrient Value	How many nutrients are present in the plant.	These are hitpoints, food value and a level gate all at once.As nutrients increase and decrease, the model is changed. If nutrients fall below zero, the plant dies.	15
Growth Stage	Whether plants are seedlings, immature or mature.	The phase of life the plant is in, it is defined / associated with a value in *Nutrients*.	
Max Nutrient Value	One cap per stage, defines the maximum number of nutrients for each stage. When the cap is exceeded, the plant grows. When it is reduced (by sickness or animal consumption) it regresses.	Numerical thresholds that gate plant growth. It is also used to determine which *growth stage* the plant is in and which model to use.	5 , 10, 15

Nutrient Tick	How often (seconds) the plant generates nutrients.		60
Can Regrow	Binary. Can be modified by Growth condition and health.		1 / 0
Regrowth Rate (Cool Down Interval)	How quickly the plant regrows after being eaten or killed in seconds. Can be modified by Growth condition and health.		60
Regrowth Potential	How many times a given plant can be regrown. Triggers automatic replanting.	The number of times a plant can regrow if it dies.	5
Compost Multiplier	Multiplier that gets applied to the Nutrient Generation Rate if the plant successfully finds Compost.		0.5

Stages of Growth:

Plants progress through three (or more, should be flexible) stages of growth based on nutrient values.
The proposed ‘default’ stages are Seedling, Immature and Mature but it would be beneficial to have a system flexible enough to support the addition or subtraction of stages.
Dinosaurs will consume nutrients from any plant regardless of the height of the plantand growth stage of dino.
All dinos can eat from any kind of plants/trees - this is represented by having low hanging branches as well as leaves on ground accessible to babies, mid-height branches for adolescents and where applicable high boughs for adults. Dinosaurs will prioritize eating mature / tall plants where possible.

And a note that the tallest adult titanosaur will still prefer the tallest most mature trees - which is already by design.

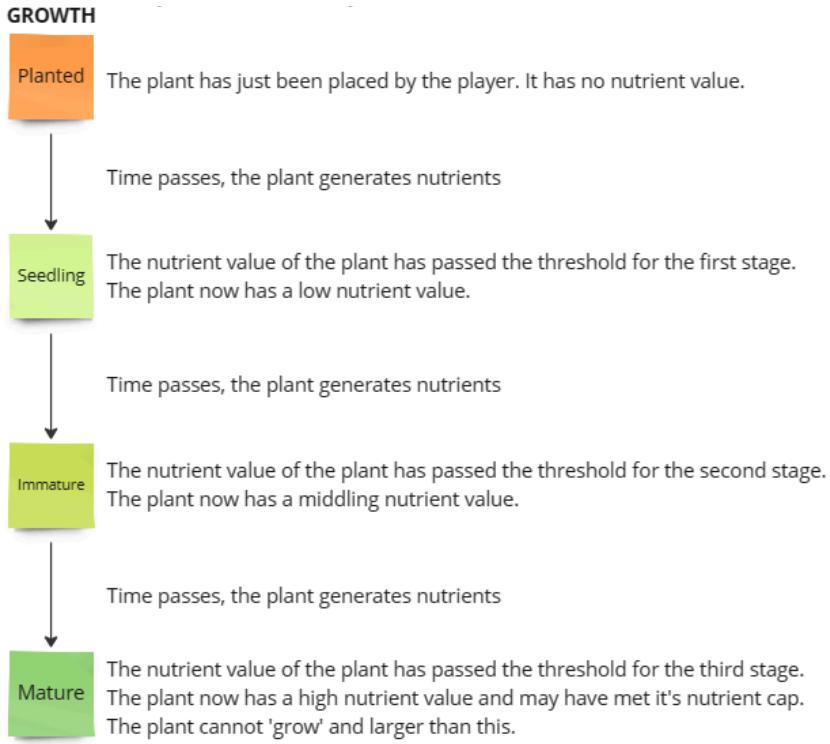
A byproduct of this preference should be that mature plants last longer.

Seedling - The plant has just been planted and growth is beginning. This is a barely edible plant that animals will only consume if they are small or desperate enough.

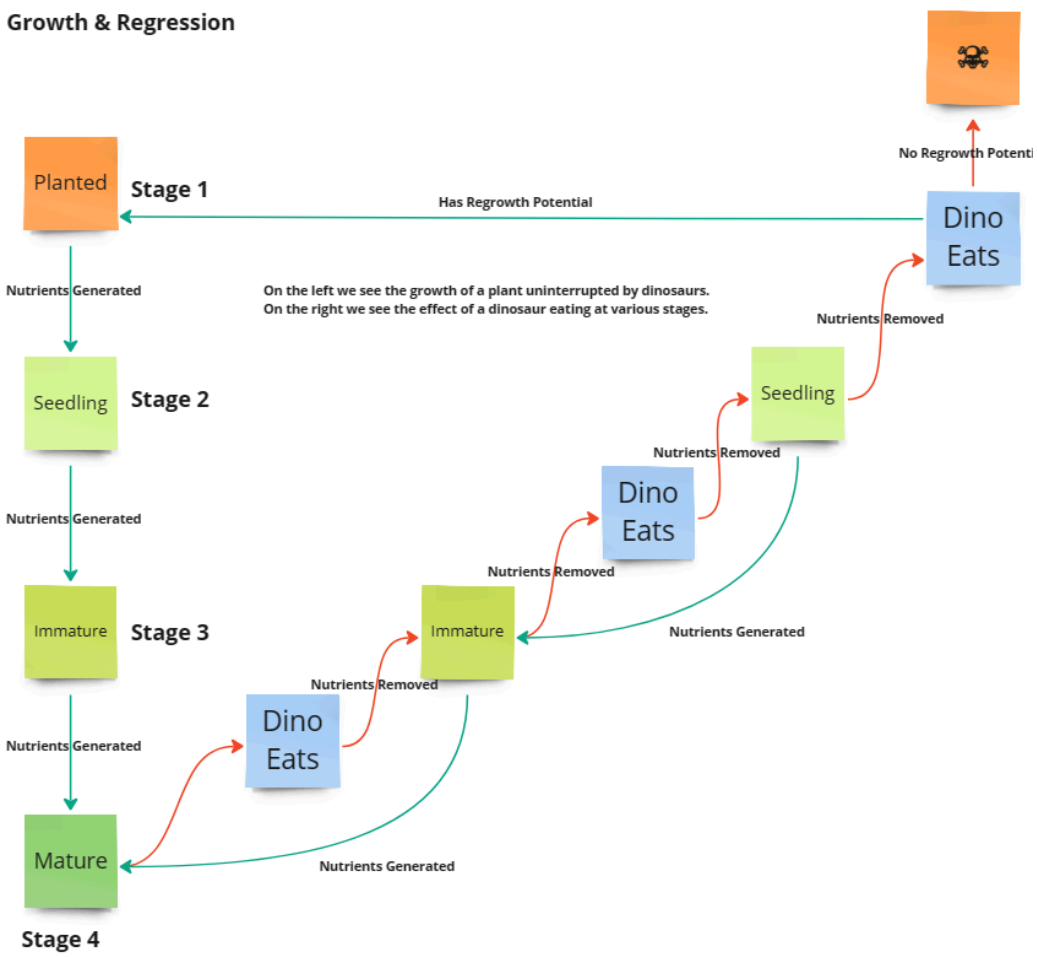
Immature - A plant in the middle of it’s growth cycle. This plant would be appetizing for a juvenile Dino. It is not enough to support most mature dinosaurs on it’s own.

Mature - A plant large enough to support a mature dinosaur. This represents the maximum possible growth potential of a plant.
Mature plants can bear fruit if they are untouched long enough.

Requirements / Considerations
We should be able to name each stage on each plant and can define different numbers of nutrients for each stage on different plants. We should be able to configure the number of growth stages on the plant quicky and easily.



[The Growth Stages of a plant.]



Parent Plants

Plants also have a Parent or Propagated status.

This is determined by a boolean (**IsParent**) This detemines which detection radius plants use to determine their likes and dislikes. Plants placed by the player are parents, plants propagated from that plant are not parent plants.

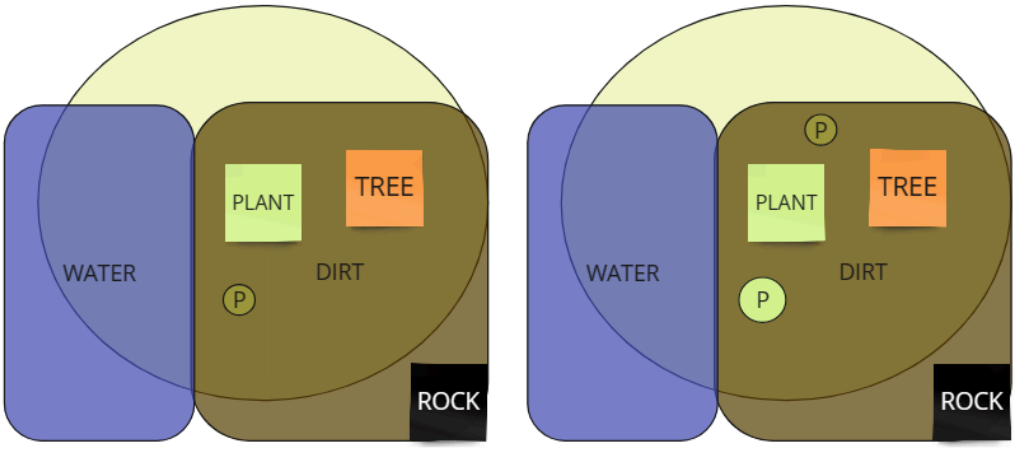
Plants placed by the player are “parents”, plants propagated from that plant are children.

Parent plants can propagate and form rest areas.

Child plants cannot propagate and have a reduced detection radius (**ChildPlantRadius**)

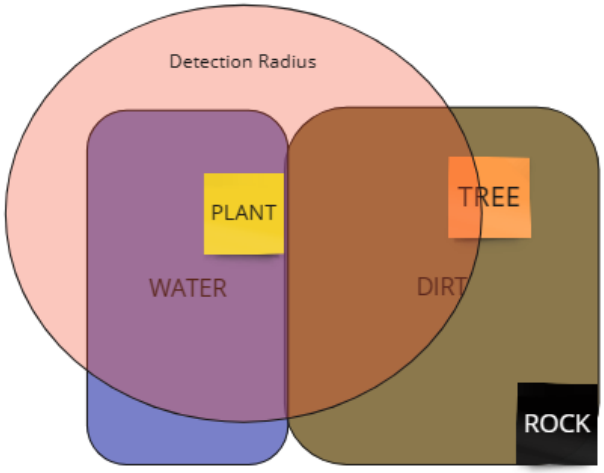
Propagation:

Plants can propagate.
Plants have different modifiers applied to their propagation rate when they are planted under different Growing Conditions.
Growing conditions are determined by whether or not the plants ‘Likes’ or ‘Dislikes’ are inside it’s detection radius. Each type of plant has its own unique desires and dislikes.
When growth conditions are not ideal, the plant may grow but will have an adverse multiplies applied to it’s rates. As an example, imagine that the plant below ‘likes’ being close to water (but not **in** water) and close to a tree. It does not like being near rocks.



Over time, a plant under ideal conditions may propagate within it’s Propagation Radius.

Above we can see that the plant is located near both of it’s likes, it has propagated two ‘child’ plants and those children are growing over time.



A plant located in poor conditions will not be able to propagate or may propagate at a reduced rate.

In the case above the plant has been located in water, which it does not like. As a result it will suffer from a penalty to propagation and will not grow children.

Variable / Condition	Explanation:	Example Value	
Propagation Enablement: 1 / 0	Status, whether or not the plant can propagate. Defined by growing conditions and health,	1	
Propagation Rate	Float, How fast propagation can occur. 1 point = 1 plant per day.	1	
Propagation Modifier Ideal	Modifiers for ideal growth condition. Affects the Propagation Rate.	+0.5	

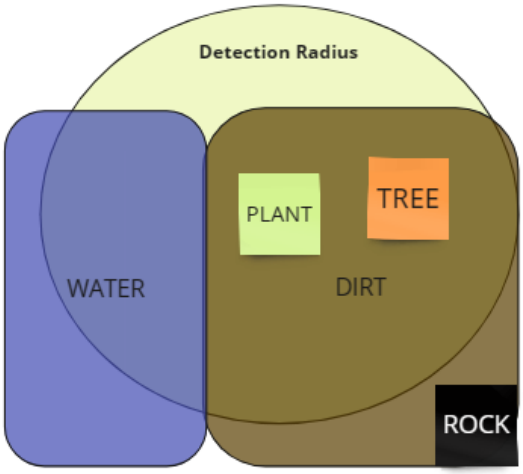
Propagation Modifier Average	Modifiers for average growth condition. Affects the Propagation Rate.	0	
Propagation Modifier Hostile	Modifiers for hostile growth condition. Affects the Propagation Rate.	-0.5	
Propagation Value	The max number of plants that can spawn in a “nearby” radius of a single parent plant. Applied to specific plants.	2	
Propagation Interval	The interval in hours before a new plant can spawn from a single parent plant.	24	
Parent Value	The original plant planted by the player, if a plant is able to propagate, it must be a parent plant in order to do so (avoids limitless propagation).	0-1	

Growing Conditions

Plants can be subjected to different growing conditions. The growing conditions needed for individual plants are defined by the individual Plant itself.

Ideal

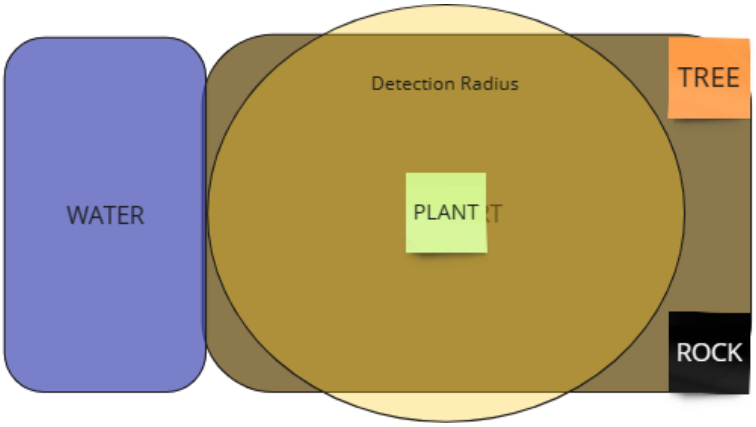
The plant is located in or near it's desired growing conditions and so it has an accellerated propagation rate. Propagation and growth are noticeably faster than under 'neutral' conditions. Proximity to fungi (and nitrogen) is often a component of ideal growing conditions.



This plant likes being close to trees (shade), water and in dirt. It does not like rocks. The tree and water are inside it's detection radius, so it is under IDEAL conditions.

Neutral

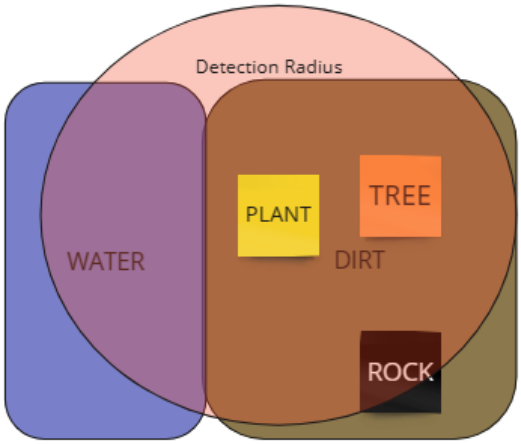
The plant is located in neutral conditions for growth - it does not sense any 'likes', nor 'dislikes' in it's detection radius.
It will be able to fully mature but may not has no modifier to propagation applied.



Tragically this plant, which loves water and trees, has been placed too far from either and so it is under 'HOSPITABLE' condition only.

Hostile

The plant is located in bad conditions for its growth, it is near things it dislikes. The plant will mature at a reduced rate and has a negative modifier applied to it's propagation rate.



Our plant has been placed too close to a rock and so it must suffer. It will have negative modifiers applied to it's growth and propagation rate.

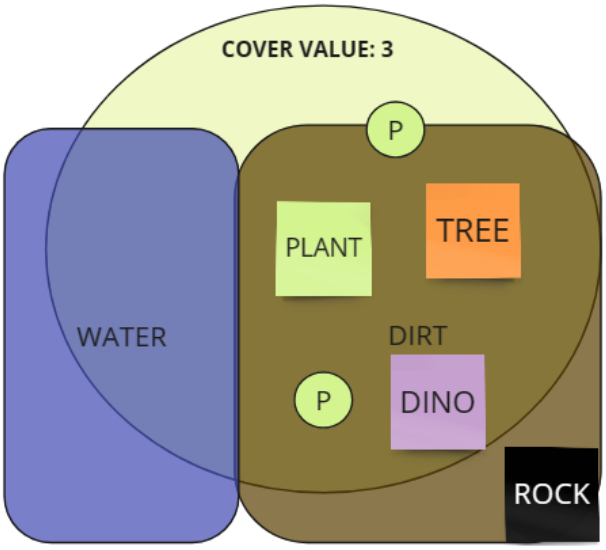
Cover

Each plant has a Cover value in points.

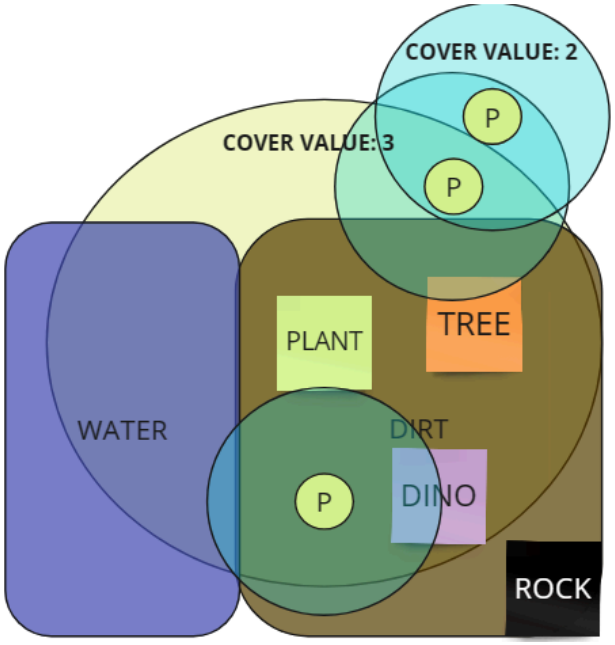
Parent plants have a larger radius than propagated plants. Propagated plants have smaller detection radius.

When multiple plants are close together (inside a single plants detection radius) this value is

added together to give the grouping a Cover Value. Dino's need a certain cover value to rest and will seek out a rest area with a high enough cover value when they need to sleep.

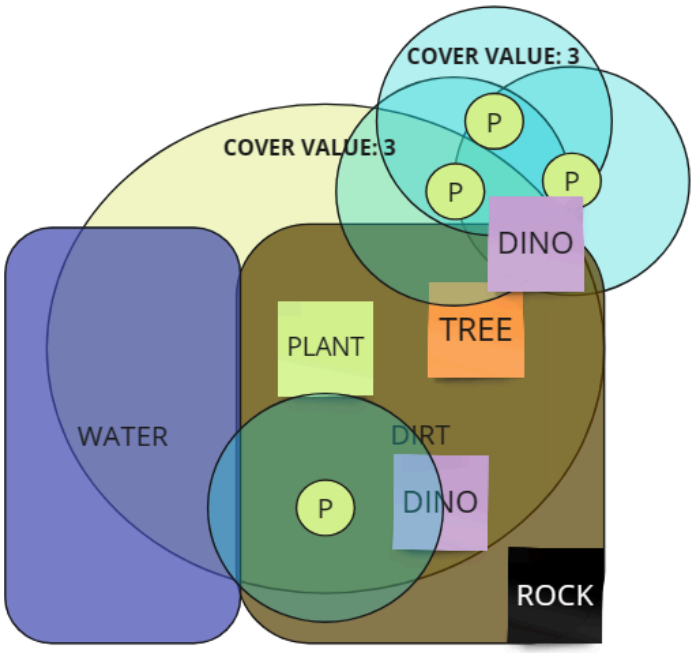


These plants are close enough to have a cover value of three, this means that a dino who needs a cover of 3 to rest can do so.



Two groupings of plants are depicted.
The propagated plants show their reduced detection radius.

Two groupings of plants are depicted.
Because the two plants at the top right are close enough to detect each-other, they are technically a cover grouping with a value of 2, but since they do not meet the Cover Grouping threshold they remain ungrouped and the dino Dino cannot rest there.



A valid group of three propagated plants

Above is an example of a valid cover grouping of three propagated plants. The cover grouping threshold has been met and the value of each plant is sufficient for the dino's to use this area for rest as well.

Variable	Description	Example Value
Cover Value	The value the plant offers in 'cover' if it is located close enough to other points.	1
~~**Cover Grouping**~~	~~How many plants need to be close enough together to be considered a rest area.~~	~~3~~
~~**IsGrouped**~~	~~Boolean that determines if plants are grouped, set to 1 if the Cover Grouping threshold is met.~~	~~0-1~~

~~**Compost Tolerance (Prototype, WiP)**~~

~~Plants dislike the presence of uncomposted dung in the environment. The plants each have an individual '**Tolerance for Compost'** value that is used to determine how much dung can be present in the environment before it is considered an unhealthy environment and the plants and dinosaurs begin to suffer.~~

~~This 'Tolerance' value of all plants in the environment is added up to get the **Compost Points Bar Max** value.~~
~~If this value is exceeded by the amount of compost points of all aggregated dung in the level, the **Fungi Disharmony Multiplier** is enabled.~~

~~**Tolerance for Compost**~~	~~Individual plant value that added up determines the Max Capacity of Compost Points the environment can tolerate.~~	~~10~~
~~**Compost Points Bar Max**~~	~~Capacity the whole environment can tolerate **Compost Points.** Determined by the Number of Plants in the environment multiplied by **Plant Tolerance for Compost** value per plant.~~	~~200~~
~~**Fungi Disharmony Multiplier (effect on Flora)**~~	~~The multiplier that gets factored into **Generation of Nutrient Interval** when **Fungi** is in Disharmony.~~	~~0.5 (e.g., all flora nutrient production rate goes from 1 to 0.5x speed~~

Death & Dormancy

Plants can die or go dormant.
Plants change state if their nutrient value drops below zero.

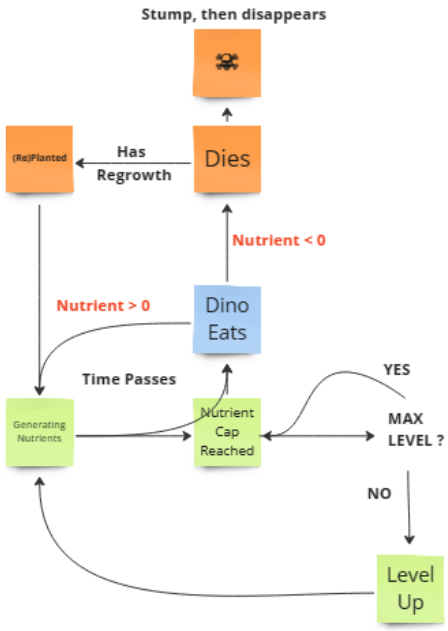
A plant with regrowth potential goes dormant for a specified time interval, a plant with no regrowth dies.
A dormant plant has no fronds or leaves but remains visually intact.
Dead plants are represented visually as a deceased, wilted version of themselves.

Dormant plants will re-generate themselves a number of times up to their maximum specified regrowth number - these plants simply revive at an appropriate growth stage.

TREES - Dormant stage is the same as the current growth stage. To exit dormancy they revert to their most recent growth stage and regrow their leaves instead of switching to the dead model.

BUSHES - Go dormant at stage 1 before reverting to their first growth stage and beginning regrowth.

Dead plants disappear after a time interval.



[The lifecycle of a plant.]

Grabbing Leaves, Fruit and Other Modular Components of Plants:

Players can interact with plants by grabbing the leaves or fruit off the plants with their hand, or if fruit with their multitool on Place/Move mode. When a leaf or fruit is grabbed off the plant, the nutrient point value of the plant goes down and is accordingly stored within the leaf or fruit.

The player's hand will have a **max amount it can store**, if it cannot grab that amount of nutrients then it will grab what is left and cause the plant to go **dormant**.

For example, a plant has a nutrient value of 10 and the player can fit 20 nutrients in their hand, therefore the player when grabbing the plant will only grab a leaf with 10 nutrients and cause the plant to go dormant.

Leaves can be grabbed **when the plant has at minimum 1 nutrient point stored**.

When grabbed, the **Nutrient Point UI ticker** will float in and show how much was removed from the original plant.

~~At **random** each day, a plant may also grow dead leaves. The player can grab these leaves off the plant to prune it, and provide a temporary multiplier to the growth rate. The dino rejects these dead leaves if fed to it~~.

The player has a couple actions they can then take with the object in their hand:

- The player can hand feed the dinosaur that object - accordingly this will increase the **Hunger Level** in the dino and also work as a Successful Needs Check that raises **Satisfaction.** See Dino/Player Interaction for additional details.
- The player can drop the object onto the ground and pick it up later. However, the leaf/fruit will disappear after a set amount of time if not picked up again.

Visually the grabbable leaves or fruit will not be attached/detached from the plant itself. It is more abstract (unless art says we can do it). When a player attempts to grab a leaf or fruit off the plant, the leaf or fruit will just appear in the players hand. The modular pieces taken off don't affect the actual mesh of the plant.

Variable	Description	Example Value
LeafNutrient	The value the leaf or fruit offers in nutrient points (calculated using **Max Plant Nutrients** divided by **Modular Pieces**)	10
MaxGrabbableNutrients	The max amount of nutrients a player can grab in their hand.	20
Timeout	The time interval, in seconds, it takes for an object dropped on the ground to disappear.	120
ProbabilityofGrowingDeadLeaves	The chance that a dead leaf will spawn on a plant.	0.1
Pruning Bonus Multiplier	When all dead leaves are removed from a plant, the plant will gain	2

	this multiplier to its nutrient growth rate for the **Pruning Bonus Interval**	
Pruning Bonus Interval	The interval in seconds in which the **Pruning Bonus Multiplier** will be applied.	120

Time Control:

Dino Points can be spent to accelerate plant growth.
See Time Control.

NOTES:

- For VS 2.0 the fern will be used to prototype these features.
  - Fern Needs : not in water, needs to be within XX distance of waters edge
  - When needs are met it multiplies it's points production by XX. If in water it reduces its point production by YY.
  - If in the right location it should also propagate (1-2 ferns appear over 2 days). Those ferns should have the same resource generation capacity as the original fern.

See a depracated list of special properties here in the Producers tab:

Enviro Stats + Features

Potential Mechanic - Maintainance.
Growing conditions can be improved for 24H by performing 'maintenance' on the plant.
Maintenance costs the player a resource or time or both.
(Maintenance is a proposed action, it is not yet defined but it is a potential sink for extra dino points. It could simply be watering or could be tied simply to the daily scanning processes somehow)

Plant Charts

Plant miro charts can be found here:

Plant Chart

Technical Documentation:

See Environment System for further description of how these plants contribute to the goal of Harmony. The following is a sample list of plants we can use for VS 2.0.

- Players use Dino Points to plant plants. It cost XX (see cost) DP to place any placeable/plant from the inventory.
- Plants generate and store their own nutrient points.
  - Plant growth is determined by number of Nutrient Points stored (e.g., a fern will appear in its Sapling/Stage 1 state if it has generated and stored between 0-5 Nutrient points, and Stage 2 stage if it has generated and stored between 6-10 Nutrient points)
  - Plants cannot die. Their models remain in the environment after harvested, and they continually produce these resources, they only get removed if the player uses the recycler on it
- Plants nutrient points storage maxed out
  - Plants have their own individual pool of nutrient points, which are capped at a max. The max varies between plant types and stage.
  - Plants can be collected/harvested (eaten by dinos) when there's at least 1 point produced/stored from the plant
- Some plants have special conditions that will multiply (or divide) the points effects (e.g., placing a fern nearby shade will multiply its Nutrient Point production).
  - For VS 2.0 we will only apply special conditions to the Fern.
- If Fungi part of the Harmony Meter is not lit/in Harmony, Flora will pause Nutrient Value Production
- Note: Plants planted in MR do not generate any points. They are merely a decorative/aesthetic modification players can perform in their MR space.

PLANT STATS	DESCRIPTION	Example Value

Nutrient Generation Rate	float (/minute), how many nutrients are generated every Nutrient tick.	1
Nutrient Value	How many nutrients are present in the plant.	15
Nutrient Cap	One cap per stage, defines the maximum number of nutrients for each stage. When the cap is exceeded, the plant grows. When it is reduced (by sickness or animal consumption) it regresses.	5 , 10, 15
Nutrient Tick	How often (seconds) the plant generates nutrients.	60
Can Regrow	Binary. Can be modified by Growth condition and health.	1 / 0
Regrowth Rate	How quickly the plant regrows after being eaten or killed in seconds. Can be modified by Growth condition and health.	60
Regrowth Potential	How many times a given plant can be regrown.	5
Growth Stage (Level)	What grow stage the Plant is at. It is essentially the plants level.	1. Regular 2. Stage 1 3. Stage 2 4. Stage 3
Propagation Enablement: 1 / 0	Status, whether or not the plant can propagate. Defined by growing conditions and health,	1
Propagation Rate	Float, How fast propagation can occur. 1 point = 1 plant per day.	1
Propagation Modifier Ideal / Average /	Modifiers for each possible growth condition. Affects the Propagation Rate.	-0.5 / +0.5

Hostile		
Propagation Value	The max number of plants that can spawn in a “nearby” radius of a single parent plant. Applied to specific plants.	2
Propagation Interval	The interval in hours before a new plant can spawn from a single parent plant.	24
Parent Value	The original plant planted by the player, if a plant is able to propagate, it must be a parent plant in order to do so (avoids limitless propagation).	Yes or No
Tolerance for Compost	Individual plant value that added up determines the Max Capacity of Compost Points the environment can tolerate.	10
Compost Points Bar Max	Capacity the whole environment can tolerate **Compost Points.** Determined by the Number of Plants in the environment multiplied by **Plant Tolerance for Compost** value per plant.	200
[Fungi Disharmony Multiplier (effect on Flora)]	The multiplier that gets factored into **Generation of Nutrient Interval** when **Fungi** is in Disharmony.	0.5 (e.g., all flora nutrient production rate goes from 1 to 0.5x speed

Available Plants for VS 2.0 and Special Conditions

DH Data (see **Plant** Data Spreadsheet - VS 2.0 Tab)

Available Plants for Rest of the Game and Special Conditions

DH Data (see Plant Data Spreadsheet - Rest of Game Tab)